Write Python programs to determine the route from your College to your Home using the following search algorithms:

- Breadth-First Search (BFS)
- Depth-First Search (DFS)
- Uniform Cost Search (UCS)
- Greedy Best-First Search
- A* Search

Include the graph used in your implementation along with edge costs and heuristic values if required.

For each algorithm, display:

- Path from College to Home
- Total path cost
- Number of nodes explored

Documentation Structure

Problem Statement

Graph Diagram

Edge Cost Table

Heuristic Table (for Greedy and A*)

Brief Description of Each Algorithm

Python Source Code

- BFS
- DFS
- UCS
- Greedy Best-First Search
- A* Search

Program Output

Screenshot or console output for each algorithm

Writeup Question Answers. (Include in documentation only)

Additional Task:

Google Maps Comparison with A* Search

Google Maps Route

Source: XYZ College

Destination: Home

Distance: 8.5 km

Estimated Time: 22 minutes

Suggested Route:

College -Junction A - Signal B - Flyover C - Home

(Insert Google Maps screenshot here.)

A* Search Output

Source: College

Destination: Home

Path Found:

College - Junction A - Signal B - Flyover C - Home

Total Cost (Distance): 8.5 km

Nodes Explored: 5

Theory Questions - **(Find answer, justify them, if needed solve on paper and add images in your document. Don't need to write answers on paper. Directly add in documentation)**

1. Which algorithm would you use for the following applications? Explain why.

- GPS Navigation
- File System Search
- Maze Solving
- Emergency Ambulance Routing
- Web Crawling

2. The graph is the same, and the goal is the same (College to Home). The only thing that changes is the data structure:

- BFS - Queue
- DFS - Stack
- UCS/Greedy Best First/A* - Priority Queue

Why do you think changing just one thing causes all the algorithms to behave differently? Explain in your own words.

3. What is the difference between:

- Shortest Path
- Least-Cost Path
- Fastest Path

4. True or False. Give reason and explanation for your answer:

- BFS always finds the least-cost path.
- DFS always finds the shortest path.
- UCS always finds the minimum-cost path.
- Greedy Search is always optimal.
- A* Search uses both cost and heuristic values.